Unit 6, Lesson 9: Applying Ratio Relationships to Percentages

Benchmark

MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.

Learning Targets

- ☐ I can make connections between ratios and percentages.
- ☐ I can use double-line diagrams, tape diagrams, and ratio tables to solve problems involving percentages.

6.9.1 Warm-Up: Estimate the Percentage

A photo of blue and pink thumb tacks is shown.

1. Based on what is shown in the picture, predict the percentage of thumb tacks that are blue by completing each statement.

   I know for sure that more than _____% of the thumb tacks are blue.

   I know for sure that less than _____% of the thumb tacks are blue.

2. My best estimate is that _____% of the thumb tacks are blue.

6.9.2 Refresher: Percentages

A ***percentage*** is a ratio expressed as a fraction of 100.

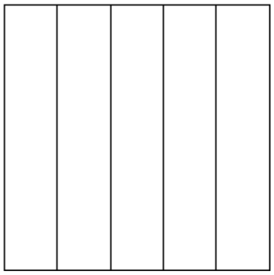
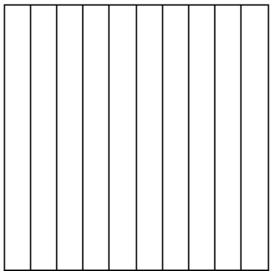
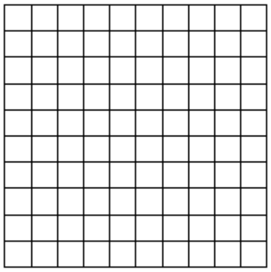
Percent $(\%)$ means "per 100," "for every 100," or "out of 100."

Work with your partner to complete the following.

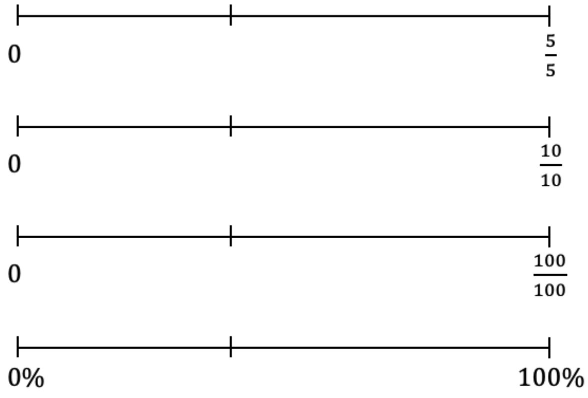
1. Each of the large squares shown are the same size.

   a. Shade $\frac{2}{5}$ of each model.

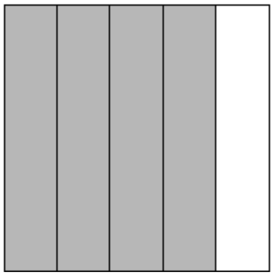
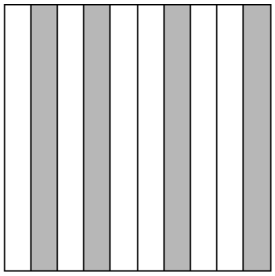
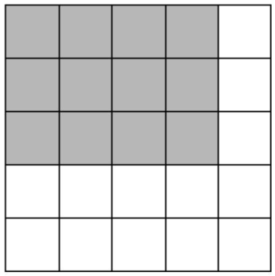
   b. Then, express the shaded part of each model as a fraction and as a percentage.

Model			
Fraction			
Percentage			

   c. Label the identified tick marks on each number line below to match the models.



2. Express the shaded part of each model as a fraction and as a percentage.

Model			
Fraction			
Percentage			

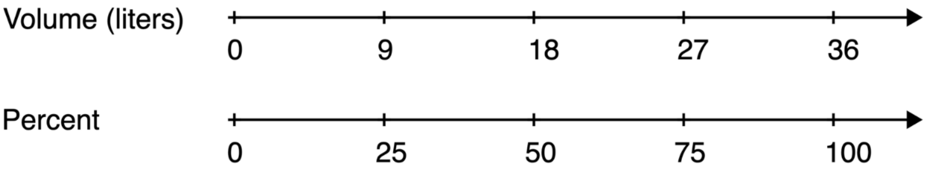
3. Complete the statements.

Percentages represent [○ part-to-part ○ part-to-whole] relationships where each percentage is a part per 100. In this case, 100% is [○ only part of the total. ○ the total, or whole, amount.]

6.9.3 Guided Instruction: Using Ratio Models to Represent Percentages

Double number lines, tape diagrams, and ratio tables can be used to model percentages similarly to other ratios. These can be helpful for determining equivalent ratios and solving problems involving percentages.

1. Consider a cooking pot that can hold 36 liters of water.
   The double number line diagram models this situation.



   a. What percentage of the pot is filled when it contains 9 liters of water?

   b. Circle where the answer to the previous question is represented on the number line diagram.

   c. How many parts were the totals on each number line, 36 liters and 100%, split into?

   d. An equivalent ratio table can also be used to model similar thinking. Determine the factor that was
   multiplied by 36 to get 9, and use it to complete the blanks below.

× ______

Volume (liters)	Percentage
36	100
9	

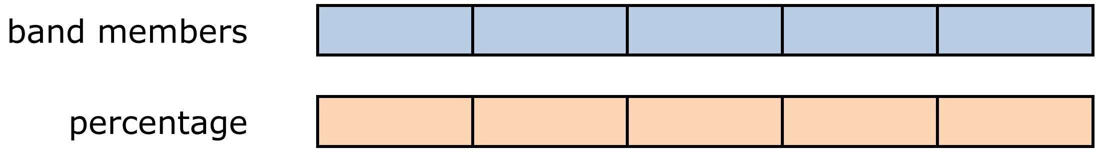
× ______

   e. Describe how the factor used to find the equivalent ratio in the table relates to how the number lines were divided in the number line diagram.

Tape diagrams can also be used to represent problems involving percentages.

2. At Millennium Middle School, there are 70 students in the school band. Of the band members, 40% are sixth graders, 20% are seventh graders, and the rest are eighth graders.

   The tape diagrams shown represent the situation.



   a. Label each small rectangle in the tape diagram of the band members with the number of students it represents.

   b. Label each small rectangle in the tape diagram of the percentage of band members with the value it represents.

   c. How many band members are sixth graders?

   d. Circle the part of the tape diagram that represents these students.

   e. How many band members are seventh graders?

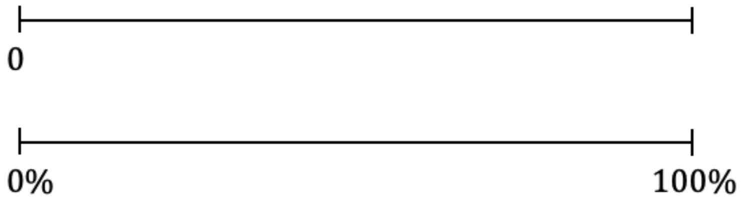
   f. Draw a rectangle around the part of the diagram that represents these students.

   g. What percentage of the band members are eighth graders?

   h. How many band members are eighth graders?

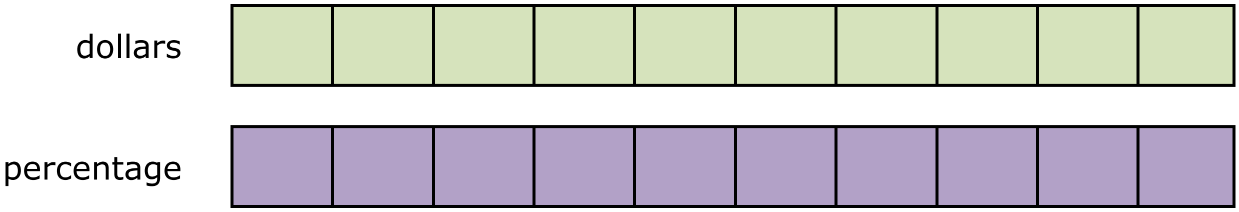
6.9.4 Guided Practice: Using Ratio Models to Represent Percentages

1. Use the number line diagram to determine what percentage of 80 is 24. Then, complete the statement.



24 is _____% of 80.

2. Use the tape diagram to determine 70% of $\$60$. Then, complete the statement.



70% of $\$60$ is $_____.

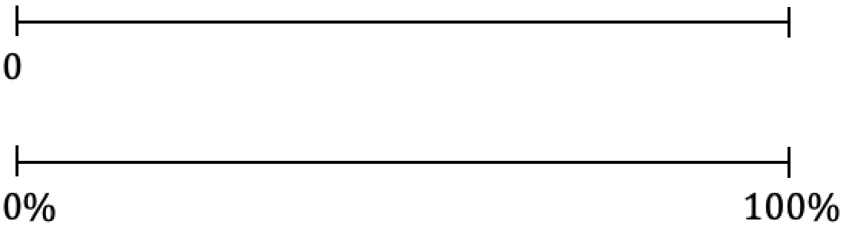
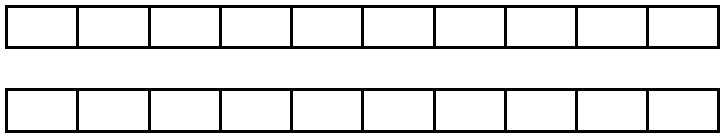
3. During a sale, a pair of jeans is on sale for 80% of its regular price. The regular price of the jeans is $\$75$. Complete the ratio table to determine the sale price. Then, complete the statement.

Price ($)	Percentage
	80
	100

80% of $\$75$ is $_____.

6.9.5 Your Turn!

1. Use the models to find the values that complete each statement. Show your work. For the last row, use the model of your choice.

<table>
<tr><th>Statement</th><th>Model</th></tr>
<tr><td>20% of 60 is _____.</td><td></td></tr>
<tr><td>_____% of 25 is 12.</td><td><table><tr><th>Value</th><th>Percentage</th></tr><tr><td></td><td>100</td></tr><tr><td></td><td></td></tr></table></td></tr>
<tr><td>40% of _____ is 28.</td><td>value
%
</td></tr>
<tr><td>16% of 125 is _____.</td><td></td></tr>
</table>

6.9.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.

1. I can make connections between ratios and percentages.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

2. I can use double-line diagrams, tape diagrams, and ratio tables to solve problems involving percentages.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

3. Write one question you still have about solving problems involving percentages after today's lesson.

